

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS Computer Science candidate (NC State, May 2026) with production experience building Java and TypeScript microservices backed by PostgreSQL, Apache Kafka, and Docker. Designed horizontally scalable, fault-tolerant systems including a transactional Java 21 + Spring Boot service with row-level locking and Testcontainers-verified concurrency correctness, and event-driven Kafka pipelines processing real-time data streams at Capgemini (20% latency reduction). Comfortable owning features end-to-end from requirements through testing and deployment.

CORE COMPETENCIES

Java + Spring Boot

Apache Kafka + Event-Driven Systems

PostgreSQL + SQL

REST API Design

Docker + Kubernetes

Microservices Architecture

Distributed Systems

CI/CD + GitHub Actions

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern | Remote

- Shipped 10+ backend features for a production AI platform -- **TypeScript, Node.js, PostgreSQL, Redis** -- across 7 deployments with 394 total commits; owned features end-to-end from requirements through monitoring.
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), role-based access control with Auth0, and real-time event streaming using Temporal.io workflow orchestration.

NC State University April 2025 - Present

Graduate Research Assistant | Raleigh, NC

- Designed a cloud-based genomic data platform (JBrowse 2 + Google Cloud Run) handling 500GB datasets for 100+ researchers worldwide.
- Collaborated on ML architecture with hyperparameter tuning and time-series analysis, improving model accuracy by 12%.

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern | Remote

- Built 5+ AI-driven automation workflows using OCR, NLP, and ML; improved health claim processing accuracy by 25%.
- Automated deployments on **AWS and GCP**; integrated Grafana monitoring dashboards; supported model validation and red-teaming for production reliability.

Capgemini Jun 2023 - Aug 2023

AI/ML Intern | Remote

- Built end-to-end distributed data pipelines using **Apache Hadoop + Spark** for batch processing and **Apache Kafka** for real-time streaming ingestion -- reduced ticket-processing latency by 20%.
- Led SDLC for an NLP-based support ticket classification system, reducing manual processing time by 15%.

PROJECTS

Inventory Reservation Service Java 21 + Spring Boot

Production-grade transactional inventory API built in **Java 21 + Spring Boot** preventing overselling via **PostgreSQL row-level locking** (SELECT FOR UPDATE) across multi-SKU reservations. Idempotent REST endpoints with RFC 9457 error responses, automated expiry via scheduled cleanup job, Flyway schema migrations, and Spring Security role-based access. Verified concurrency correctness with Testcontainers integration tests -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict. Multi-stage Docker build, GitHub Actions CI.

Java 21 · Spring Boot 3.5 · PostgreSQL · Flyway · Spring Data JPA · Spring Security · Testcontainers · Docker · GitHub Actions | github.com/Aditya-k24/InventoryReservationService

VectorLift -- Semantic Search + Ranking Engine Kafka + Docker Compose

Production-grade search system on MS MARCO: BM25 (Elasticsearch), dense bi-encoder retrieval (PyTorch fine-tuned), and cross-encoder re-ranker. **Kafka-driven indexing pipeline** for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI serving API with per-stage latency tracking; full-stack deployment with **PostgreSQL, Redis, Prometheus, Grafana, Docker Compose**. Evaluated 6 pipeline configurations with NDCG@10, MRR@10, MAP and paired bootstrap significance testing.

Python · Kafka · FastAPI · PostgreSQL · Redis · Elasticsearch · Qdrant · Prometheus · Grafana · Docker Compose | github.com/Aditya-k24/VectorLift

AlgoPulse -- Agentic AI Orchestration Platform Kafka + Temporal

Event-driven AI orchestration platform scaled to 2,000 async inference requests/second. Identified API-polling bottleneck under load and pivoted to Supabase Edge Functions streaming with **Temporal.io** for workflow state management and **Kafka** for fault-tolerant state synchronization -- zero state loss under load.

TypeScript · Node.js · React · Supabase Edge Functions · Temporal.io · Kafka | github.com/Aditya-k24/AlgoPulse

PulseRank -- Real-Time Leaderboard & Activity Feed Distributed Data

Event-driven leaderboard platform: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for feed document projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads. Validated multi-store consistency with Testcontainers; benchmarked hot paths with k6; shipped observability with OpenTelemetry + Prometheus and CI/CD with GitHub Actions.

TypeScript · NestJS · MySQL · MongoDB · Redis · Docker · OpenTelemetry · Prometheus · GitHub Actions | github.com/Aditya-k24/PulseRank

EDUCATION

MS Computer Science -- NC State University	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms	
BTech Computer Science -- Mumbai University	Aug 2020 - May 2024
CGPA: 9.32/10 Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, OS, Machine Learning, Software Engineering	

SKILLS

- Languages:** Java, Python, TypeScript, JavaScript, SQL, C, R
- Backend:** Spring Boot, Spring Data JPA, Spring Security, Flyway, Node.js, NestJS, FastAPI, REST API
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant
- Messaging/Pipelines:** Apache Kafka, Apache Spark, Hadoop, Temporal.io, Testcontainers
- DevOps/Cloud:** Docker, Kubernetes, GitHub Actions, AWS, GCP, Prometheus, Grafana, Terraform